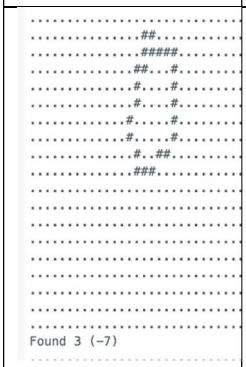
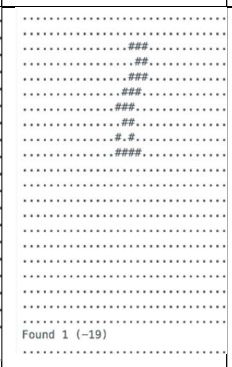
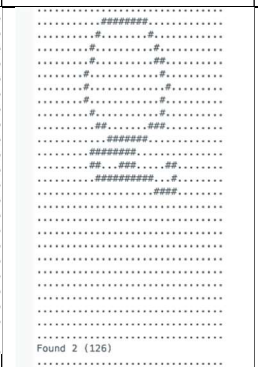
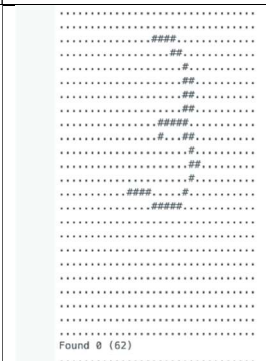
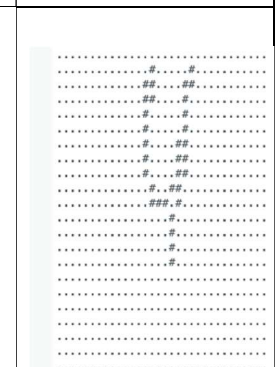
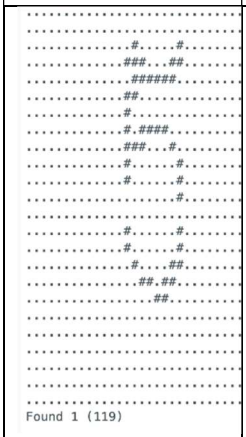
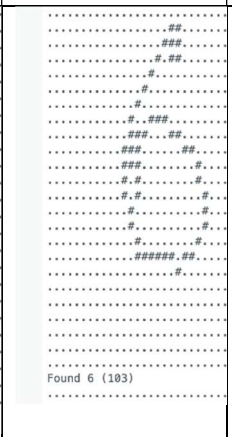
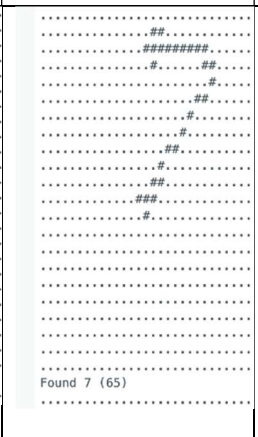
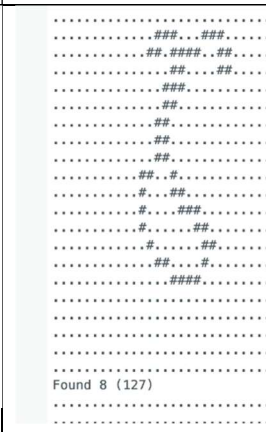
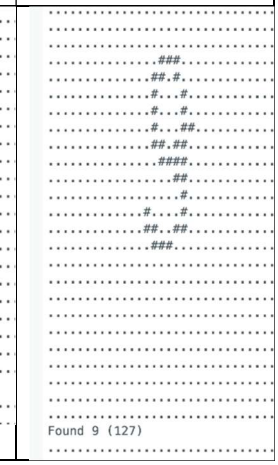
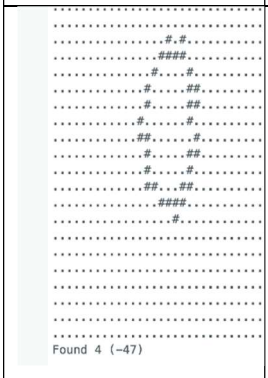
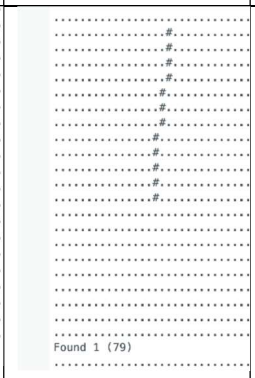
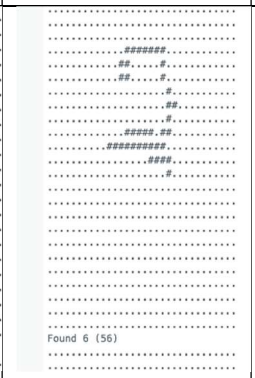
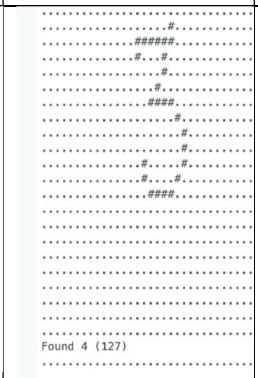
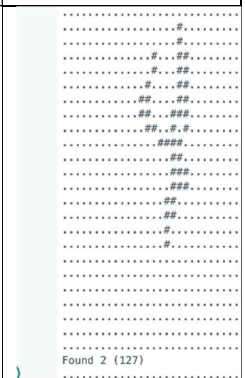
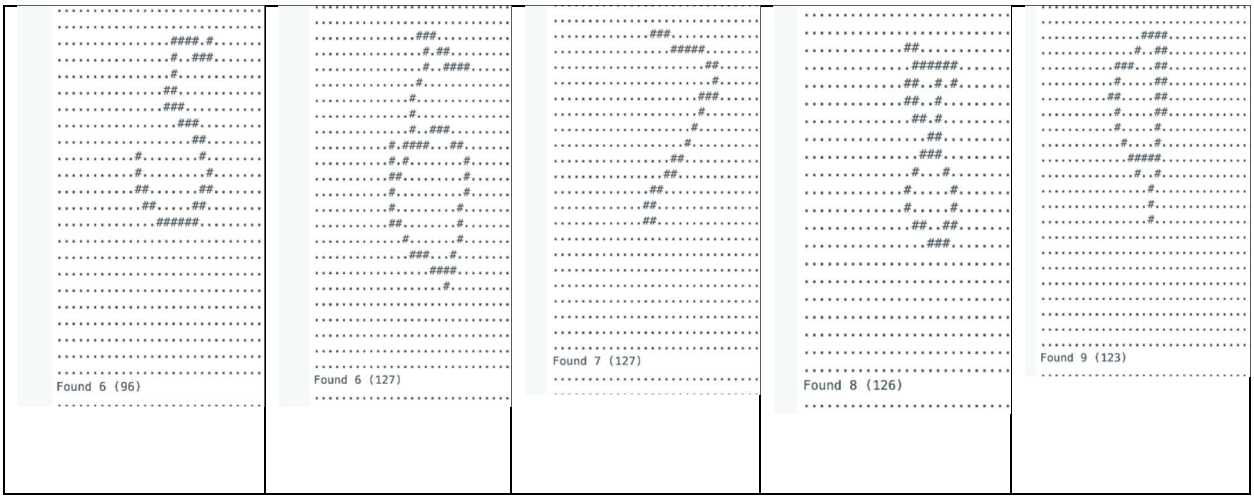
EE 446 - Lab 7: Wizard Magic Wand

Baseline Model:

0	1	2	3	4
				
5	**6**	**7**	**8**	**9**
				

Finetuned Model:

0	1	2	3	4
				
5	**6**	**7**	**8**	**9**



Discussion and Reflection:

The finetuned model was better at recognizing certain numbers, however it was still incorrect sometimes. The model had a hard time detecting anything circular, often getting it messed up with zero, six, two, or three. Despite the model having errors, the errors were less frequent than the baseline and also got better at detecting specific numbers like 7, 8, and 9.

Overall, both models struggled. This was due to our data collection. The website was difficult to use and would delete our data when turned into the json file or would mess up the labels we wrote. For example, we got over 20 samples of 1, but when we opened the JSON file there were zero. This made the data hard to match, and why the model struggled to detect the difference between a lot of the more circular numbers.

Fine tuning definitely did help, since it made the model more accurate to picking up our gestures. Once we figured out the best technique for holding the board and doing the motions it got much better at detecting the correct number. If we had better data and less difficulties with the JSON file, the fine-tuned model would perform much better than how it did in the lab.